Chaz Cornwall

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EDUCATION

Michigan Technological University | College of Engineering Houghton, MI

Ph.D. Electrical Engineering, 4.0 GPA May 2026

- Dissertation: "The Excess Path Length Distribution: A Model for Sample-based Path Planners"
- Created novel stochastic models to improve convergence and uncertainty quantification of sample-based path planners

Utah State University | College of Engineering Logan, UT

M.S. Electrical Engineering, 4.0 GPA Dec 2022

- Thesis: "Event-based Obstacle Detection with Commercial Lidar"
- Researched event-based LiDAR processing, behavior cloning, and state estimation for obstacle detection

B.S. Electrical Engineering, Minor: Spanish, 3.95 GPA May 2021

- Developed embedded C++ code for a self-steering baby walker (Senior Design)

RESEARCH EXPERIENCE

Air Force Institute of Technology Dayton, OH

Air Force Science and Technology Fellow (administered by National Research Council) Jun 2026– Present

- Applying uncertainty quantification to state estimation and AI feature detection
- Supporting PhD students by advising on research direction, abstract writing, and technical approaches

Michigan Technological University Houghton, MI

Graduate Research Assistant Jan 2023 – May 2026

- Developed white papers for path planning optimization, HMI, and off-road perception research

TEACHING EXPERIENCE

Air Force Institute of Technology Dayton, OH

Co-instructor for Special Topic Course: Integrity for Visual Systems Summer 2026

- Developed and presented lectures on classic and state-of-the-art aviation integrity frameworks
- Assisted students plan and debug final projects (certifiable optimization and integrity calculations for vision-based navigation)

Michigan Technological University Houghton, MI

Guest Lecturer for EE5531: Introduction to Robotics Jan 2026

- Designed two lectures, a homework assignment, and a lab for a module on optimal state estimation
- Delivered two 50-minute lectures

Graduate Teaching Assistant for GM/SAE AutoDrive Challenge II Jan 2023 – May 2026

- Acted as Lead TA by guiding team technical approach to meet competition requirements given constraints (~5 TAs)
- Advised the mapping/planning and perception sub-teams (~15 students)
- Developed baseline object tracking, path planning, and state estimation systems that facilitate parallel development and unit testing
- Oversaw autonomous vehicle software architecture

PROFESSIONAL EXPERIENCE

Autonomous Solutions, INC Petersboro, UT

Perception Engineer Sept 2021 – Dec 2022

- Developed existence estimation algorithm for RADAR detections
- Created neural network design and evaluation framework with PyTorch and TensorFlow
- Improved optical character recognition pipeline for trailer number detection

Perception Engineering Intern Mar 2020 – Sept 2021

- Oversaw preparation of the obstacle detection system for a customer demo
- Developed a robust RADAR software driver in ROS 2 to handle dropped network connections

Space Dynamics Laboratory Logan, UT

Electrical Engineering Assistant Oct 2018 – Mar 2020

- Generated testing procedures for small satellite assembly and integration
- Developed analog circuitry and power conversion for sun sensor test fixtures on small satellites' ADCS
- Participated in decisions that involved all student employees (Student Scholar, Vice Chair)

PUBLICATIONS/PRESENTATIONS

Journals

- Chaz Cornwall, Clark Taylor. "The Consistency of Covariance Estimates in Symmetric Maximum Likelihood Estimation". IEEE Transactions on Automatic Control. 2026. In preparation.
- Chaz Cornwall, Logan Schexnaydre, Jeremy Bos. "Sample-based Path Planner Hyperparameter Tuning with Cost Model Acceleration". IEEE TASE. 2026. In preparation.
- Chaz Cornwall, Logan Schexnaydre, Ian Mattson, Casey Majhor, Jeremy Bos. "Can We Get There Faster: Tuning Sample-Based Path Planners". IEEE RAL. 2026.
- Chaz Cornwall and Scott Budge. "Obstacle detection using range-difference events". Optical Engineering. SPIE. 2022.

Conferences

- Aaron DeLuca and Chaz Cornwall. "Creating Visual Navigation Protection Levels Using a Mini-Operational Design Domain (ODD) Design". ION ITM. 2026. In preparation.
- Clark Taylor and Chaz Cornwall. "S4: Speedy Solution Separation using Subadditivity in Multiple Fault Exclusion". ION ITM. 2026. In preparation.
- Chaz Cornwall, L. Schexnaydre, I. Mattson, Y. Yiming, D. Babcock, N. Petersen, J. P. Bos, D. Robinette. "Caerus Polaris: A Minimally-viable Autonomous Vehicle". SAE Non-event Technical Paper. 2026. In preparation.
- Logan Schexnaydre, Ian Mattson, Chaz Cornwall, and Jeremy Bos. "Modeling the Effects of Residual Snow Clutter on Position Estimation and Vehicle Energy Efficiency". IEEE International Workshop on Metrology for Automotive. 2026.
- Chaz Cornwall and Jeremy Bos. "Solving the Right Problem with Multi-Robot Formations". SAE WCX. 2026.
- Chaz Cornwall and Scott Budge. "Biology-inspired LiDAR-centric obstacle detection". Autonomous Systems: Sensors, Processing and Security for Ground, Air, Sea and Space Vehicle and Infrastructure. SPIE. 2022.
- Chaz Cornwall and Scott Budge. "A neuromorphic approach to LiDAR point cloud processing". Autonomous Systems: Sensors, Processing, and Security for Vehicles and Infrastructure. SPIE. 2021.

Presentations/Posters

- Chaz Cornwall. "The Excess Path Length Distribution". AFRL Discover Symposium. 2026.
- Chaz Cornwall. "Are We There Yet: Tuning Sample-Based Path Planners". Michigan Tech Graduate Research Colloquium. 2025.
- Chaz Cornwall. "Wire-cutting Mechanism for Aeroboom Deployment". Utah Conference on Undergraduate Research. 2019.

AWARDS/FELLOWSHIPS

Air Force Science and Technology Fellowship (AF STFP) - $200000 Jun 2026 – Jun 2028

- Submitted proposal on conservative sensor fusion using sample-based approaches (approved and recommended by the National Research Council)
- Work on autonomous aerial refueling and vision-aided navigation at the Air Force Institute of Technology

Top 10% Student Reviews Jan 2025 – Apr 2025

- Received a top-10% student rating from TA sections

W. Brent & Beverly J. Robinson Scholarship - $5000 Sep 2021 – Aug 2022

- Exhibited leadership while undergraduate student

Utah NASA Space Grant Consortium Fellowship - $5000 Oct 2018 – Aug 2019

- Researched electrical circuits for inflatable boom deployment onboard a CubeSat
- Designed/Built/Tested prototype circuit boards to be flown on high altitude balloons

Presidential Scholarship – Full Tuition Aug 2017 – May 2021

SERVICE

Peer Reviewer

- IEEE RAL, IROS
- ION Navigation

IEEE Robotics and Automation Society (RAS) Student Chapter Houghton, MI

Chair Nov 2025 – April 2026

- Create the chapter by interacting with faculty, students, and submitting a petition to IEEE
- Organize an initial meeting to determine future activities and inform potential members
- Planned path planning activity and coordinated with the IEEE Student Branch

Get Away Special Team, USU Department of Physics Logan, UT

Team Coordinator May 2019 – Mar 2020

- Acted as Program Manager/Chief Engineer of the Get Away Special Team managing ~25 students
- Organized operations to reach CubeSat mission critical deadlines
- Received Utah NASA Space Grant Consortium Fellowship

Electrical Team Lead May 2018 – April 2019

- Prepared for Get Away Special Passive Attitude Control System (GASPACS) CubeSat mission PDR
- Participated in CubeSat electrical schematic design

SKILLS / OTHER

- Proficient in C, C++, Python, MATLAB, ROS, ROS 2, PyTorch, TensorFlow, Linux (Ubuntu)